

VDA 7.1

Hugo Lladró Prats, Inés Mang Román, Ahmet Meriç Kızıltas

June 4, 2026

a)

A parametrization of the sphere using latitude ϕ , and longitude λ is given by:

$$\mathbf{p}(\phi, \lambda) = (\cos \phi \sin \lambda, \cos \phi \cos \lambda, \sin \phi)$$

With $(\phi, \lambda) \in (-\frac{\pi}{2}, \frac{\pi}{2}) \times [0, 2\pi]$, because ϕ determines the height of the parallel, and λ determines the position around it.

b)

Using this representation we have that:

$$\begin{aligned}\frac{d\mathbf{p}}{d\phi} &= (-\sin \phi \sin \lambda, -\sin \phi \cos \lambda, \cos \phi) \\ \frac{d\mathbf{p}}{d\lambda} &= (\cos \phi \cos \lambda, -\cos \phi \sin \lambda, 0) \\ \frac{d\mathbf{p}}{d\phi} \times \frac{d\mathbf{p}}{d\lambda} &= \begin{vmatrix} i & j & k \\ -\sin \phi \sin \lambda & -\sin \phi \cos \lambda & \cos \phi \\ \cos \phi \cos \lambda & -\cos \phi \sin \lambda & 0 \end{vmatrix} = \\ &= \cos \phi \begin{vmatrix} i & j & k \\ -\sin \phi \sin \lambda & -\sin \phi \cos \lambda & \cos \phi \\ \cos \lambda & -\sin \lambda & 0 \end{vmatrix} = \\ &= \cos \phi (\cos \phi \sin \lambda, \cos \phi \cos \lambda, \sin \phi) \\ \left\| \frac{d\mathbf{p}}{d\phi} \times \frac{d\mathbf{p}}{d\lambda} \right\| &= \cos \phi \sqrt{\cos^2 \phi \sin^2 \lambda + \cos^2 \phi \cos^2 \lambda + \sin^2 \phi} = \cos \phi\end{aligned}$$

c)

Using the components $x = \lambda$, $y = \sin \phi$ over $(-\frac{\pi}{2}, \frac{\pi}{2}) \times [0, 2\pi]$, we notate $\psi = (x, y)$:

$$\begin{aligned}\frac{d\psi}{d\phi} &= (0, \cos \phi) \\ \frac{d\psi}{d\lambda} &= (1, 0) \\ \frac{d\psi}{d\phi} \wedge \frac{d\psi}{d\lambda} &= -\cos \phi \\ \left| \frac{d\psi}{d\phi} \wedge \frac{d\psi}{d\lambda} \right| &= \cos \phi\end{aligned}$$

We check that the area elements of both parametrizations coincide, so the map projection preserves area.